Capstone Project Final Report

ECOM30002/90002 Econometrics 2 - Semester 2, 2025

The Gender Dividend: How Does Gender Specific Educational Investments Differently Impact Household Economic Welfare in Burkina Faso?

Abstract:

This report examines how gender causally affects marginal returns to education in the context of economic welfare, using household consumption as a proxy. Our analysis finds that females experience substantially larger returns to education than males, recognising the greater economic access women gain from education. While robustness checks conjunctively with extensions are performed to support the use of two stage least squares method, we are aware that there may still be endogeneity concerns. Findings highlight the necessity for targeted policy approaches that foster greater education for females.

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1. Introduction

1.1 Motivation

Education stands as the cornerstone of human-centered development. World Bank studies across Africa demonstrate educational investments yield economic returns that surpass most forms of capital investment (UNICEF, 1990, p.45). In a regional context, Burkina Faso presents a particularly vital challenge, ranking among the lowest performers on human development indicators in sub-Saharan Africa (Browne et al., 1991), with persistent lags in school completion rates, especially for women (World Bank, 1990), that constrain economic participation and labor market access. While educational attainment remains imperative for everyone, evidence points toward prioritising women's education (Browne et al., 1991). This focus is grounded in compelling data: education emerges as the single most significant factor explaining gender disparities in Burkina Faso, accounting for 41% of overall gender inequality (Batana et al., 2013).

Female education is a catalyst for household transformation. Its effects radiate immensely through better hygiene practices, improved nutrition, greater household health outcomes, dramatic reductions in child marriage, and lower birth rates (Flagg et al., 2013). These benefits are synergistic, each creating compound returns that change household intergenerational economic trajectories (UN Women, 2020). This sparks our curiosity: how much does gender specific educational investments affect household welfare, using household consumption as a proxy.

1.2 Mechanism & Key Relationship

The key relationship is between gender-education interactions and household consumption. Women in Burkina Faso confront systematic structural barriers including constrained labor market access, restricted property rights, and persistent wage discrimination. A women's education attainment significantly influences household wealth, with percent change in wealth score being significant only in households where women had primary or secondary education (Miller et al., 2017). We posit that higher educational attainment among women serves as a mechanism to dismantle these binding constraints, generating disproportionately higher returns, relative to men, as measured through household consumption per capita.

1.3 Contributions of research

Our analysis will help define the impact of gender specific educational investments on the welfare in the household. Understanding this can potentially help policy makers and education heads design a schooling environment that is more accessible and supports gender-equal education, whilst also, overcoming household notions that investments on female education is unnecessary due to economic barriers that women face in the region.

2. Data Description

Our analysis includes the following variables summarised in Table 1:

Statistic	Mean	St. Dev.	Median	Min	Max
Log Consumption	12.995	0.744	12.966	10.579	16.390
Education (Years)	8.321	3.004	6	1	16
Interaction (Female x Education)	3.557	4.368	0	0	16
Female	0.441	0.497	0	0	1
Urban	0.791	0.407	1	0	1
Log HH Size	1.776	0.628	1.792	0.000	3.932
Religion Majority	0.579	0.494	1	0	1
Employment Returns	0.196	0.397	0	0	1
Maternity Hospital Access	0.677	0.468	1	0	1
Elementary School Access	0.980	0.140	1	0	1
Middle School Access	0.817	0.386	1	0	1
High School Access	0.737	0.440	1	0	1

Note: number of observations is 65,015.

Table 1: Descriptive Statistics on Education and Household Consumption

Our dataset consists of 65,015 observations from the 2018 Burkina Faso Harmonized Survey on Household Living Conditions (EHCVM), after excluding observations with incomplete records on education level.

Our dependent variable is individual-level log consumption, defined as the natural log of household total consumption (measured in Burkina Faso CFA francs) divided by household size. The total consumption aggregate follows standard LSMS/EHCVM practice and includes food (purchased, home-produced, in-kind), non-food, services, rent, health, education, and other items over their respective recall periods.

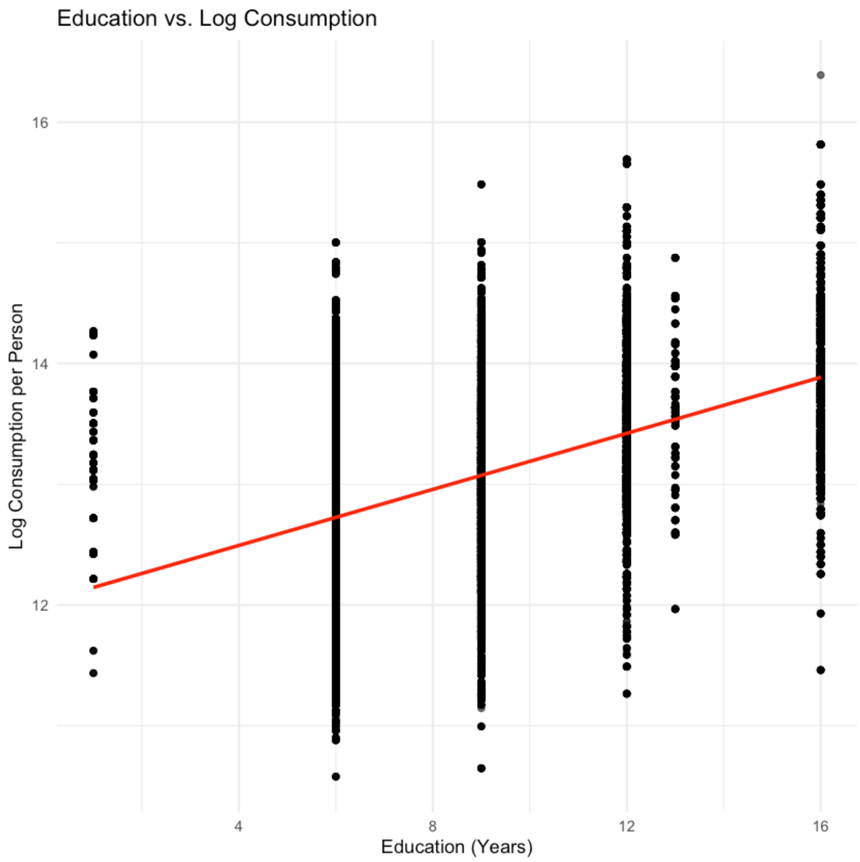

Figure 1: Explanatory Variable Education(years) vs. Dependent Variable Log Consumption

Our explanatory variable, years of education (Figure 1), is a measure of completed schooling years derived from the categorical education-level variable. Each education level was converted into its corresponding number of schooling years (e.g., Primary = 6 years, Higher Education = 16 years) in accordance with the structure of Burkina Faso's education system. Furthermore, to examine gender differences in the impact of education on welfare, we constructed an interaction term(Female × Education) that captures the differential effect of education between men and women.

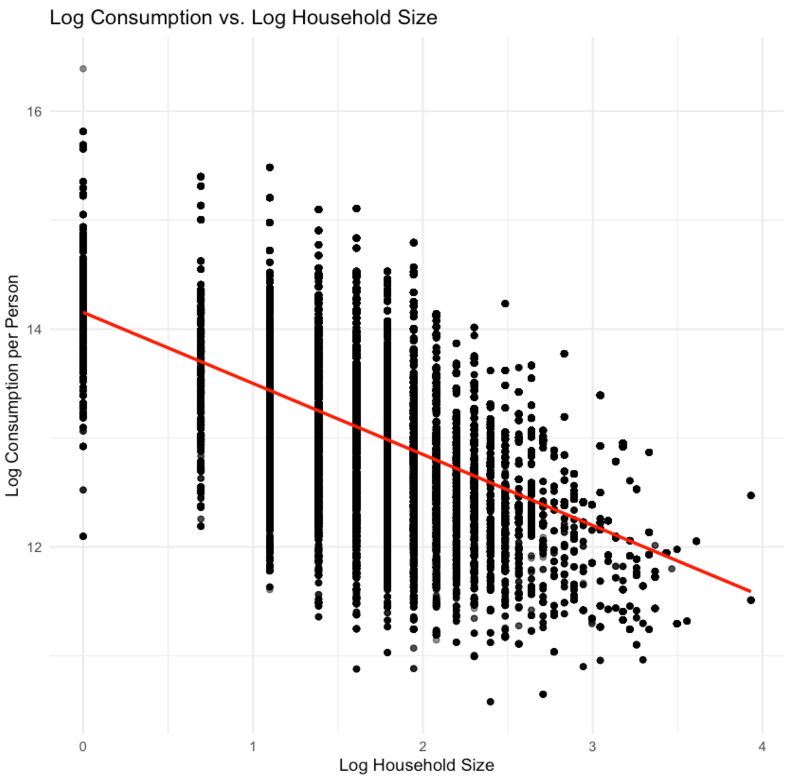

Figure 2: Control Variable Log Consumption vs. Dependent Variable Log Consumption

Figure 2 illustrates that consumption per person tends to decrease as the number of household members increases, consistent with Agyepong, L. et al. (2024). We also add in other control variables, which will be shown in the methodology part.

3. Methodology

3.1. OLS Method

Variable	Expected Relationship & Rationale	Expected Direction
Education (Years)	Individuals with more education are expected to have higher earning potential and make more informed financial decisions. We assume this would lead to higher household consumption.	+
Interaction (Female × Edu)	It is hypothesized that education may unlock greater economic opportunities for females than for males, especially in contexts with existing gender gaps. Therefore, the marginal return to education is assumed to be higher for females.	+
Female	Holding education constant, females may face wage gaps or have different labor market participation rates, which could result in lower average consumption compared to males.	−
Urban	Urban areas typically offer better job opportunities, higher wages, and more access to markets, all of which are expected to lead to higher household consumption compared to rural areas.	+
Log(Household Size)	For a given level of household resources, a larger household size means that consumption must be divided among more people, leading to lower per capita consumption.	−

Table 2. OLS Variables Explanation

3.2. 2SLS Method

Endogenous Variable(s)	Rationale for Instrument Relevance	Expected Strength
Education (Years)	The exact 4 IVs are listed in the equation later. Community-level access to basic infrastructure like maternity hospitals and elementary schools during childhood is expected to be a strong predictor of final educational attainment. Better access reduces the cost and difficulty of starting and continuing education.	Strong ($F > 10$)
Interaction (Female × Edu)	The interaction of these access instruments with gender (Access × Female) is expected to strongly predict the interaction term. This is because the impact of school access on educational attainment may be different for girls than for boys, allowing the model to identify the differential returns.	Strong ($F > 10$)

Table 3. 2SLS Variables Explanation

To obtain a causal estimate of the effect of education on consumption, we employ an Instrumental Variable (IV) approach. This method is appropriate because of potential endogeneity concerns in the education variable. Specifically, we suspect omitted variable bias arising from unobserved factors such as individual ability, gender-specific characteristics, parental background, and motivation, which may simultaneously influence both education and consumption outcomes. Moreover, reverse causality is likely, as wealthier households may be able to afford more years of schooling for their members. Finally, there may be measurement error in reported education levels, given that respondents could misstate or approximate their years of schooling.

First Stage

$$EducationYears_i = \alpha_6 + \alpha_7 MaternityAccess_i + \alpha_8 MaternityAccess_i : MaternityAccess_i +$$
$$\alpha_9 ElementaryAccess_i + \alpha_{10} ElementaryAccess_i : Female_i + \gamma_{11} Female_i + \gamma_{12} Urban_i + \gamma_{13} Log\ HouseholdSize_i$$

$$Female_i\ EducationYears_i = \alpha_{14} + \alpha_{15} MaternityAccess_i + \alpha_{16} MaternityAccess_i : MaternityAccess_i +$$
$$\alpha_{17} ElementaryAccess_i + \alpha_{18} ElementaryAccess_i : Female_i + \gamma_{19} Female_i + \gamma_{20} Urban_i + \gamma_{21} Log\ HouseholdSize_i$$

Second Stage

$$log\ Consumption_i = \beta_0 + \beta_1 EducationYears_{i_{hat}} + \beta_2 EducationYears * Female_{i_{hat}} +$$
$$\gamma_3 Female_i + \gamma_4 Urban_i + \gamma_5 Log\ HouseholdSize_i$$

We address endogeneity by using local infrastructure as instruments—specifically, indicators for maternity (hospital), elementary access in the community, and each of their interaction terms with females.

These IVs satisfy the two core requirements.

- Relevance. Greater accessibility (to local infrastructure) reduces travel and opportunity costs, which raises early enrollment and encourages continued schooling. Consequently, the proximity/availability of schools strongly predicts years of schooling—especially for girls—by lowering both direct and indirect costs.

- Exogeneity: local infrastructure placement is largely determined by administrative and historical factors that pre-date current household consumption, so conditional on region fixed effects, urban/rural status, cohort, and community amenities (e.g., roads, markets, health posts), local infrastructure access should affect household consumption only through its impact on education.

4. Results

	Dependent variable:		
	Log Consumption per Person		
	Model 1: OLS Baseline	Model 3: OLS Full	Model 5: 2SLS Final
	(1)	(2)	(3)
Education (Years)	0.1128***	0.0738***	0.4045***
	(0.0009)	(0.0009)	(0.0199)
Interaction (Female × Education)	0.0139***	0.0027	0.2740***
	(0.0006)	(0.0015)	(0.0231)
Female (1=Female)		0.0784***	-1.9390***
		(0.0133)	(0.1858)
Urban (1=Urban)		0.5069***	-0.0745*
		(0.0051)	(0.0334)
Log Household Size		-0.4926***	-0.0541*
		(0.0033)	(0.0247)
Constant	12.0073***	12.8111***	9.6658***
	(0.0077)	(0.0114)	(0.1876)
Observations	65,015	65,015	65,015
R2	0.2264	0.4943	-2.3979
Adjusted R2	0.2264	0.4943	-2.3982
Residual Std. Error	0.6540 (df = 65012)	0.5287 (df = 65009)	1.3706 (df = 65009)
F Statistic	9,514.3760*** (df = 2; 65012)	12,710.3400*** (df = 5; 65009)	

Table 4: Regression Result of OLS Baseline, OLS with Control Variable, 2SLS Model

4.1. OLS Results and Interpretation

The full OLS specification, Model 3, is displayed as the second column in table 4. This provides a baseline before addressing endogeneity. The coefficient for "Education (Years)" is 0.0738 and is highly significant. This suggests that for males, each additional year of schooling is associated with a 7.38% increase in per capita consumption. The "Interaction (Female × Education)" coefficient is 0.0027 but is not statistically significant. This indicates that there is no detectable difference in returns between genders. For females, the total estimated return is 7.65% (0.0738 + 0.0027) per year.

4.2. 2SLS Results and Interpretation

The final 2SLS specification (Model 5) is displayed as the third column in table 4. This corrects for endogeneity and provides causal estimates. The "Education (Years)" coefficient is 0.4045 and is highly significant, indicating that for males, each additional year of schooling causally increases per capita consumption by 40.45%. The "Interaction (Female ×

Education)" coefficient is 0.2740 and is also highly significant. This positive and significant interaction reveals that returns to education are substantially higher for females. The total causal return for females is 67.85% (0.4045 + 0.2740) for each additional year of schooling. This demonstrates a strong, gender-differentiated impact of education on economic welfare.

5. Robustness Checks

5.1. Robustness Check for OLS

	Dependent variable:		
	Log Consumption per Person		
	Model 1	Model 2 Add Controls	Model 3 Final OLS
	(1)	(2)	(3)
Education (Years)	0.113***	0.074***	0.074***
	(0.001)	(0.001)	(0.001)
Female x Education	0.014***	0.003	0.003
	(0.001)	(0.002)	(0.002)
Female (1=Female)		0.078***	0.078***
		(0.013)	(0.013)
Urban (1=Urban)		0.506***	0.507***
		(0.005)	(0.005)
Log Household Size		-0.494***	-0.493***
		(0.003)	(0.003)
Religion Majority (1=Muslim)		0.010*	
		(0.004)	
Employment Returns (1=Yes)		0.002	
		(0.005)	
Constant	12.007***	12.806***	12.811***
	(0.008)	(0.012)	(0.011)
Observations	65,015	65,015	65,015
R2	0.226	0.494	0.494
Adjusted R2	0.226	0.494	0.494
Residual Std. Error	0.654 (df = 65012)	0.529 (df = 65007)	0.529 (df = 65009)
F Statistic	9,514.376*** (df = 2; 65012)	9,080.096*** (df = 7; 65007)	12,710.340*** (df = 5; 65009)
Note:			*p<0.05; **p<0.01; ***p<0.001

Table 5: Robust OLS Regression Results of Gender-Differentiated Returns to Education on Consumption

Based on t-tests with robust standard errors, the control variables Religion_Majority and Employment_returns were found to be statistically insignificant in Model 2. They were subsequently removed to create Model 3, our OLS Model.

```
	studentized Breusch-Pagan test

data:  model3
BP = 443.09, df = 5, p-value < 2.2e-16
```

Table 6: Heteroskedasticity Test (Breusch-Pagan) for Model 3

```
	Jarque Bera Test

data:  residuals(model3)
X-squared = 126.15, df = 2, p-value < 2.2e-16
```

Table 7: Normality Test (Jarque-Bera) for Model 3

```
	RESET test

data:  model3
RESET = 156.88, df1 = 2, df2 = 65007, p-value < 2.2e-16
```

Table 8: Model Specification Test (RESET) for Model 3

We then run a more comprehensive diagnosis on model 3. This revealed several critical econometric issues. The Breusch-Pagan test in table 6 detected significant heteroskedasticity, confirming the necessity of using robust standard errors for valid inference, which we applied throughout the analysis. While a normality test in table 7 showed that residuals were not perfectly normal, the large sample size (n > 5,000) allows for reliance on the Central Limit Theorem for asymptotic validity. Most importantly, the Ramsey RESET test in table 8 indicated significant model misspecification. This result strongly suggests the presence of endogeneity, where education is likely correlated with unobserved factors in the error term.

5.2. Robustness Check for 2SLS

```
Coefficient Comparisons (OLS vs 2SLS Model 5):
Education coefficient:       OLS = 0.0738, 2SLS = 0.4045 (448.0% change)
Interaction coefficient:     OLS = 0.0027, 2SLS = 0.2740 (10211.3% change)

Maximum percentage difference: 10211.3%
```

Table 9: Endogeneity Test (Wu-Hausman Test)for necessity of 2SLS

Firstly, the endogeneity test on table 9 reveals a substantial difference between the OLS and 2SLS coefficients. It displays that endogeneity was a significant issue and justifying the necessity of the instrumental variable approach.

```
--- F-TEST FOR INSTRUMENT STRENGTH (Education) ---

Linear hypothesis test:
MaternityHospitalAccess = 0
ElementarySchoolAccess = 0
MaternityHospitalAccess:female = 0
ElementarySchoolAccess:female = 0

Model 1: restricted model
Model 2: education_years ~ MaternityHospitalAccess + ElementarySchoolAccess +
    MaternityHospitalAccess:female + ElementarySchoolAccess:female +
    female + urban + household_size

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1  65011 523758
2  65007 518958  4    4800.4 150.33 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 10: IV Relevance Test (First Stage F-Test) for EDUCATION_YEARS

```
--- F-TEST FOR INSTRUMENT STRENGTH (Interaction) ---

Linear hypothesis test:
MaternityHospitalAccess = 0
ElementarySchoolAccess = 0
MaternityHospitalAccess:female = 0
ElementarySchoolAccess:female = 0

Model 1: restricted model
Model 2: interaction_female_education ~ MaternityHospitalAccess + ElementarySchoolAccess +
    MaternityHospitalAccess:female + ElementarySchoolAccess:female +
    female + urban + household_size

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1  65011 193521
2  65007 189975  4    3545.6 303.31 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 11: IV Relevance Test (First Stage F-Test) for FEMALE X EDUCATION

Table 10 and table 11 shows that the first-stage IV is relevant to the 2 endogenous variables. This is indicated by F-statistics exceeding the standard threshold of 10 for both regressions.

```
Linear hypothesis test:
education_years = 0
interaction_female_education = 0

Model 1: restricted model
Model 2: log_consumption ~ education_years + interaction_female_education +
    female + urban + household_size | MaternityHospitalAccess +
    ElementarySchoolAccess + MaternityHospitalAccess:female +
    ElementarySchoolAccess:female + female + urban + household_size

  Res.Df Df  Chisq Pr(>Chisq)
1  65011
2  65009  2 508.24  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 12: Joint Significance Test (Wald Test) for 2 Endogenous Variables

The joint significance Wald test in table 12 $\chi^2 = 508$ rejects the null hypothesis that these education-related coefficients are jointly zero. This confirms that education and its female interaction term is collectively significant.

```
Instruments: 4 (Maternity, Elementary + 2 female interactions)
Endogenous variables: 2 (education_years, interaction_female_education)
Over-identification degrees of freedom: 2
      df1       df2 statistic   p-value
 2.000000        NA  3.544031  0.169990
```

Table 13: Over-Identification Restriction Test (Sargan J-test) for IV Exogeneity

The Sargan J-test in table 13 for over-identifying restrictions passes with a p-value of 0.16. Since this is greater than 0.05, we cannot reject the null hypothesis of instrument exogeneity and therefore maintain the belief that all instrument variables are exogenous.

Collectively, these tests validate Model 5 as a robust specification for estimating the causal effects of education.

5.3. Robustness Check for invalid 2SLS (Exploration Process)

```
Configuration:
- Base instruments: 4 (MaternityHospitalAccess, ElementarySchoolAccess, MiddleSchoolAccess,
HighSchoolAccess)
- Total instruments (including interactions): 8
- Endogenous variables: 2 (education_years, interaction_female_education)
- Over-identification degrees of freedom: 6

SARGAN J-TEST RESULTS:
Sample size: 65015
R² from auxiliary regression: 0.000648
Sargan J-statistic: 42.0985
Degrees of freedom: 6
P-value: 0.0000
```

Table 14: Over-Identification Restriction Test (Sargan J-test) for IV Exogeneity on Invalid 2SLS Model

When building the 2SLS Model, we did begin with a different set of instrument variables. That is, we used a total of 8 instrument variables that include 4 base IVs (maternity hospital access, elementary school access, middle school access, high school access) and their female interaction term. In this setting, from table 14, we find p-value smaller than 0.05 and we reject null hypothesis. This shows that some instruments may be invalid. We then tried many subsets of this setting and found that the two instrument variables, middle school access and high school access may be endogenous. To explain endogenity, we believe high School Access might directly affect consumption through better local infrastructure, economic development in the area, and network effects beyond just education.

After deploying different subsets of the instrument variable set, we end up choosing the current set of instrument variables that is both relevant and exogenous.

6. Conclusion

Conclusively, we find there is strong and statistically significant evidence that for each additional year of education, females experience 27.40% larger returns to education than males. Specifically, each additional year of education increases household consumption by 40.45% for male household heads and 67.85% for females. Hence, aligning with the introductory hypothesis that being female is associated with higher returns to education, relative to men. To mitigate OVB, we enact several control variables, and thereafter due to concerns of reverse causality and unobserved confounders between education and consumption, we use the 2SLS IV approach using school access and school access x female.

Results were supported by Wu-Hausman, Wald, F, and Sargan J tests and are robust. However, due to concerns of endogeneity remaining in the final model through unobservable factors that affect gender, and gender*education such as life span, child marriage, sexism, and health we are not satisfied with a causal interpretation of our coefficients. Thus, while we conclude that our findings are merely descriptive, not causal, we find strong evidence that females experience larger returns to education than men. There is apparent motivation for further research and targeted policies that support further education for females to ensure economic prosperity for households and the country.

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